

Python Code Questions

1. Revision of Basics of Python

1.1 Python / Basics

1. Write a Python program to print "Hello, Python!".
2. Write a Python program to display your name and school name.
3. Write a Python program to print three lines using print().
4. Write a Python program to print the result of $5 + 3$, $10 - 2$, $4 * 6$, and $20 / 4$.
5. Write a Python program that uses comments and prints a message.

1.2 Fundamental Syntax of Python

1. Write a Python program using correct indentation to display whether a number is positive or negative.
2. Write a Python program with single-line and multi-line comments.
3. Write a Python program that shows Python case sensitivity by using variables with different names.
4. Write a Python program to show a simple code block inside if.

2. Input and Output

2.1 Input Statement

1. Write a Python program to take a user's name as input and print a greeting.
2. Write a Python program to take two numbers as input and display them.
3. Write a Python program to take age as input and print it.

2.2 Output Statement

1. Write a Python program to print your name, class, and section.
2. Write a Python program to print the sum of two numbers.
3. Write a Python program to print a formatted message using print().

2.3 Type Conversion with Input

1. Write a Python program to take a number as input and convert it into integer.
2. Write a Python program to take a decimal number as input and convert it to float.
3. Write a Python program to take two numbers as input, convert them to integers, and find the sum.
4. Write a Python program to take a number as a string and convert it to integer.

3. Data Types and Variables

3.1 Variables

1. Write a Python program to declare variables for name, age, and marks.
2. Write a Python program to swap two variables.
3. Write a Python program to assign one value to multiple variables.
4. Write a Python program to show valid and invalid variable names.

3.2 Data Types

A. Numeric Data Type

1. Write a Python program using integer values.
2. Write a Python program using float values.
3. Write a Python program to add two complex numbers.

B. Text Data Type

1. Write a Python program to store and print a string.
2. Write a Python program to combine two strings.

C. Boolean Data Type

1. Write a Python program to check whether a number is greater than 10.
2. Write a Python program to print the result of a Boolean expression.

D. Sequence Data Type

List

1. Write a Python program to create a list of five fruits.
2. Write a Python program to access the first and last element of a list.

Tuple

1. Write a Python program to create a tuple of three numbers.
2. Write a Python program to print items from a tuple.

E. Set Data Type

1. Write a Python program to create a set of unique numbers.
2. Write a Python program to add an element to a set.

F. Mapping Data Type

1. Write a Python program to create a dictionary of student details.

2. Write a Python program to print the keys and values of a dictionary.

G. None Type

1. Write a Python program to assign None to a variable and print it.

4. Operators and Expressions

4.1 Arithmetic Operators

1. Write a Python program to use all arithmetic operators with two numbers.
2. Write a Python program to find quotient and remainder of two numbers.
3. Write a Python program to calculate power using **.

4.2 Relational Operators

1. Write a Python program to compare two numbers using all comparison operators.
2. Write a Python program to check whether a number is equal to 50.
3. Write a Python program to find whether one number is greater than another.

4.3 Logical Operators

1. Write a Python program to check if a number is between 10 and 20 using and.
2. Write a Python program to check whether a number is less than 5 or greater than 15 using or.
3. Write a Python program to use not with a Boolean value.

4.4 Assignment Operators

1. Write a Python program to demonstrate =, +=, -=, * =.
2. Write a Python program to update a variable using assignment operators.
3. Write a Python program to calculate a value using //=, %= and **=.

5. String Formatting

5.1 Format Specifiers

1. Write a Python program using %d to print an integer.
2. Write a Python program using %f to print a decimal number.
3. Write a Python program using %.2f to print a number up to two decimal places.
4. Write a Python program using %s to print a string.
5. Write a Python program using %x and %o.

5.2 f-Strings

1. Write a Python program using f-string to print your name and age.

2. Write a Python program using f-string to display the sum of two numbers.
3. Write a Python program using f-string to print a number up to two decimal places.
4. Write a Python program using f-string with an expression inside {}.
5. Write a Python program using f-string to access values from a dictionary.

6. Conditional Statements

6.1 if Statement

1. Write a Python program to check whether a number is positive.
2. Write a Python program to check whether a student passed if marks are above 40.

6.2 if-else Statement

1. Write a Python program to check whether a number is even or odd.
2. Write a Python program to check whether a person is adult or not.

6.3 if-elif-else Statement

1. Write a Python program to display grade based on marks.
2. Write a Python program to display whether a number is positive, negative, or zero.
3. Write a Python program to check whether a year is leap year.

6.4 Nested if

1. Write a Python program to check whether a number is positive and even.
2. Write a Python program to check eligibility for voting and driving.

7. Python Collections

7.1 String

1. Write a Python program to print each character of a string.
2. Write a Python program to reverse a string.
3. Write a Python program to count the length of a string.

7.2 List

1. Write a Python program to create a list and print all elements.
2. Write a Python program to add an item to a list.
3. Write a Python program to remove an item from a list.
4. Write a Python program to access list elements by index.

7.3 Tuple

1. Write a Python program to create a tuple and print all items.
2. Write a Python program to access tuple elements.
3. Write a Python program to count items in a tuple.

7.4 Dictionary

1. Write a Python program to create a dictionary of student information.
2. Write a Python program to print only keys of a dictionary.
3. Write a Python program to print only values of a dictionary.
4. Write a Python program to print key-value pairs of a dictionary.

8. Range Function

8.1 range()

1. Write a Python program using range(stop) to print numbers from 0 to 4.
2. Write a Python program using range(start, stop) to print numbers from 1 to 5.
3. Write a Python program using range(start, stop, step) to print even numbers from 2 to 20.
4. Write a Python program to print a multiplication table using range().

9. Loops

9.1 for Loop

1. Write a Python program to print numbers from 1 to 10 using for.
2. Write a Python program to print the multiplication table of 7.
3. Write a Python program to traverse a list using for.
4. Write a Python program to print each character of a string using for.

9.2 while Loop

1. Write a Python program to print numbers from 1 to 5 using while.
2. Write a Python program to find the sum of numbers from 1 to 10 using while.
3. Write a Python program to keep taking input until the user enters 0.

10. Loop Control Statements

10.1 break

1. Write a Python program to stop a loop when number 5 is found.
2. Write a Python program to find the first even number in a list using break.

10.2 continue

1. Write a Python program to skip number 3 in a loop.
2. Write a Python program to print numbers from 1 to 10 but skip multiples of 3.

10.3 else with Loop

1. Write a Python program to print “Loop completed” using else with for.
2. Write a Python program to search a number in a list using loop and else.

11. Nested Loops

11.1 Definition-based programs

1. Write a Python program to print a 3×3 grid of numbers.
2. Write a Python program to print a multiplication table using nested loop.
3. Write a Python program to print a square pattern using stars.
4. Write a Python program to print a triangle pattern using nested loops.

12. Built-in Library Functions

12.1 String Functions

1. Write a Python program using upper().
2. Write a Python program using lower().
3. Write a Python program using center().
4. Write a Python program using len().

12.2 Mathematical Functions

1. Write a Python program using sum().
2. Write a Python program using pow().
3. Write a Python program using round().
4. Write a Python program using abs().
5. Write a Python program using sqrt().
6. Write a Python program using int().

13. Functions in Python

13.1 Function

1. Write a Python program to define and call a function.
2. Write a Python program to create a function that prints your name.
3. Write a Python program to create a function that adds two numbers.

13.2 Types of Functions

1. Write a Python program using a built-in function like print() or len().
2. Write a Python program using a user-defined function to display a message.

13.3 Advantages of User-defined Functions

These are mostly theory, but code can be asked like:

1. Write a Python program using a function to avoid repeated code.
2. Write a Python program using two function calls to show reusability.

13.4 Defining a Function

1. Write a Python program to define a function and use parameters.
2. Write a Python program to define a function that calculates area of a rectangle.
3. Write a Python program to define a function that calculates simple interest.

13.5 Calling a Function

1. Write a Python program to define a function and call it.
2. Write a Python program to call a function twice.

14. Scope of Functions

14.1 Scope

1. Write a Python program showing a local variable.
2. Write a Python program showing a global variable.

14.2 Local Scope

1. Write a Python program to define a variable inside a function and print it inside the function.
2. Write a Python program to show that a local variable cannot be used outside the function.

14.2 Global Scope

1. Write a Python program to use a global variable inside and outside a function.
2. Write a Python program to modify a global variable using global.

14.3 Difference Between Local and Global Scope

1. Write a Python program to show both local and global variables with the same name.

15. Function Parameters

15.1 Parameters

1. Write a Python program with a function that takes one parameter.
2. Write a Python program with a function that takes two parameters.

15.2 Types of Parameters

Required / Positional Parameter

1. Write a Python program to add two numbers using positional parameters.
2. Write a Python program to display full name using two parameters.

Default Parameter

1. Write a Python program to greet a person with a default name.
2. Write a Python program to calculate area using default length and breadth.

Keyword Parameter

1. Write a Python program to display student details using keyword arguments.
2. Write a Python program to display book title and price using keyword arguments.

Variable Length Parameter

1. Write a Python program to add any number of numbers using *args.
2. Write a Python program to print any number of names using *args.
3. Write a Python program to calculate total marks using *args.

****kwargs**

1. Write a Python program to display student information using **kwargs.
2. Write a Python program to print product details using **kwargs.

16. Function Arguments

16.1 Function Arguments

1. Write a Python program to pass a number as an argument.
2. Write a Python program to pass a string as an argument.

16.2 Types of Arguments Passed

Number

1. Write a function that takes a number and prints its square.

String

1. Write a function that takes a string and prints it in uppercase.

Boolean

1. Write a function that takes True or False and displays a message.

List

1. Write a function that takes a list of marks and prints them.

Tuple

1. Write a function that takes a tuple of subject names and prints them.

Dictionary

1. Write a function that takes a dictionary and prints its keys and values.

Variable

1. Write a function that takes a variable and prints its value.

Expression

1. Write a function that takes an expression like $5 + 3$ as argument.

Another Function

1. Write a Python program where one function's return value is passed to another function.

17. Function Return Type

17.1 return Statement

1. Write a Python program where a function returns a number.
2. Write a Python program where a function returns a string.
3. Write a Python program where a function returns a Boolean value.

17.2 Returning Values

1. Write a Python program to return the sum of two numbers.
2. Write a Python program to return the area of a square.
3. Write a Python program to return a list of numbers.
4. Write a Python program to return a tuple of coordinates.
5. Write a Python program to return a dictionary of student details.

18. Passing Concepts in Functions

18.1 Mutable Objects

1. Write a Python program to modify a list passed to a function.
2. Write a Python program to modify a dictionary passed to a function.
3. Write a Python program to modify a set passed to a function.

18.2 Immutable Objects

1. Write a Python program to show that integer passed to a function does not change outside.
2. Write a Python program to show that a string passed to a function does not change outside.
3. Write a Python program to show that a tuple passed to a function remains unchanged.

19. Dry Run and Program Tracing

19.1 Dry Run

1. Write the output of a sequential program and show dry run.
2. Write the output of a conditional program and show dry run.
3. Write the output of a looping program and show dry run.
4. Write the output of a function program and show dry run.

19.2 Sequential Flow

1. Trace a program with variable assignment and arithmetic operations.

19.3 Conditional Structures

1. Trace an if program.
2. Trace an if-else program.
3. Trace an if-elif-else program.
4. Trace a nested if program.

19.4 Loop Structures

1. Trace a for loop program.

2. Trace a while loop program.
3. Trace a nested loop program.

19.5 User-defined Functions

1. Trace a function returning a value.
2. Trace a function calculating sum or square.
3. Trace a function that returns result to another function.

20. Modules, Packages and Libraries

20.1 Module

1. Write a Python program to import a module and use one function from it.
2. Write a Python program to use a user-defined module.

20.2 from Keyword

1. Write a Python program using from math import sqrt.
2. Write a Python program using from random import randint.

20.3 Package

1. Write a Python program showing how a package is used.
2. Write a program structure that uses a package folder.

20.4 Library

1. Write a Python program using any built-in library.
2. Write a Python program using a third-party library.

21. Importing Libraries

21.1 Import Entire Library

1. Write a Python program using import math.

21.2 Import Specific Function

1. Write a Python program using from math import sqrt.

21.3 Import All Functions

1. Write a Python program using from math import *.

22. Popular Python Libraries

22.1 Math Library

1. Write a Python program using `sqrt()`.
2. Write a Python program using `ceil()`.
3. Write a Python program using `floor()`.
4. Write a Python program using `factorial()`.
5. Write a Python program to print the value of `pi`.

22.2 Random Library

1. Write a Python program using `random()`.
2. Write a Python program using `randint()`.
3. Write a Python program using `randrange()`.
4. Write a Python program using `choice()`.
5. Write a Python program using `shuffle()`.
6. Write a Python program using `uniform()`.

22.3 Pandas Library

1. Write a Python program to create a `DataFrame`.
2. Write a Python program to read a CSV file using `pandas`.
3. Write a Python program to display the first five rows of a `DataFrame`.

22.4 Turtle Library

1. Write a Python program to draw a square using `turtle`.
2. Write a Python program to draw a triangle using `turtle`.
3. Write a Python program to draw a circle using `turtle`.

22.5 Matplotlib Library

1. Write a Python program to plot a line graph.
2. Write a Python program to plot a bar chart.
3. Write a Python program to plot a pie chart.

23. Graphics Using Turtle

1. Write a Python program to draw a line using `turtle`.
2. Write a Python program to draw a square using `turtle` commands.
3. Write a Python program to draw a rectangle using `turtle` commands.
4. Write a Python program to draw a triangle using `turtle` commands.
5. Write a Python program to draw a circle using `turtle` commands.
6. Write a Python program to change `turtle` color and speed.

7. Write a Python program to use penup() and pendown().
8. Write a Python program to draw a shape with a turtle icon.

24. Error Handling

24.1 Errors and Exceptions

1. Write a Python program that produces a syntax error example.
2. Write a Python program that produces a runtime error example.
3. Write a Python program that produces a logical error example.

24.2 Types of Errors

These are mostly theory, but code-based questions may include identifying errors in given programs.

25. Try-Except

25.1 try-except Block

1. Write a Python program to handle division by zero using try-except.
2. Write a Python program to handle invalid input using try-except.

25.2 Multiple Except Blocks

1. Write a Python program to catch ValueError and ZeroDivisionError.

25.3 Else with try-except

1. Write a Python program using try-except-else.

25.4 Finally Block

1. Write a Python program using try-except-finally.

25.5 Blank Input Error

1. Write a Python program to check for blank input and display an error message.

26. File Handling

26.1 File Handling

1. Write a Python program to explain file handling through code.
2. Write a Python program to create and store data in a file.
3. Write a Python program to read a file and display its contents.

26.2 File Modes

1. Write a Python program showing file opening in read mode.
2. Write a Python program showing file opening in write mode.
3. Write a Python program showing file opening in append mode.

27. Basic File Operations

Opening a File

1. Write a Python program to open a file.
2. Write a Python program to create a new file.

Reading a File

1. Write a Python program to read the contents of a file using read().

Writing a File

1. Write a Python program to write text into a file using write().

Closing a File

1. Write a Python program to close a file after opening it.

28. Reading Data from File

1. Write a Python program to read a file line by line using readline().
2. Write a Python program to read all lines of a file using readlines().
3. Write a Python program to print the number of lines in a file.

29. CSV File Handling

29.1 CSV File

1. Write a Python program to create a CSV file.
2. Write a Python program to read data from a CSV file.

29.2 CSV Standard Library

1. Write a Python program to write data to a CSV file using csv module.
2. Write a Python program to read data from a CSV file using csv module.

30. Pandas File Handling

30.1 Introduction

1. Write a Python program to read a CSV file using pandas.
2. Write a Python program to read an Excel file using pandas.

30.2 Pandas File Functions

1. Write a Python program using `pd.read_csv()`.
2. Write a Python program using `pd.read_excel()`.
3. Write a Python program using `pd.read_json()`.
4. Write a Python program using `pd.read_html()`.
5. Write a Python program to create a DataFrame.
6. Write a Python program to save a DataFrame to CSV.
7. Write a Python program to save a DataFrame to Excel.

31. CSV Handling Using Pandas

1. Write a Python program to create a CSV file using pandas.
2. Write a Python program to read a CSV file using pandas.
3. Write a Python program to display head and tail of a CSV file.
4. Write a Python program to print shape and columns of a DataFrame.
5. Write a Python program to calculate the mean of numeric columns.
6. Write a Python program to display statistical summary using `describe()`.

32. Data Visualization

1. Write a Python program to visualize a data set.
2. Write a Python program to plot marks of students.
3. Write a Python program to plot monthly sales data.

33. Matplotlib

33.1 Introduction

1. Write a Python program to create a plot using matplotlib.
2. Write a Python program to display a chart title and axis labels.

33.2 Common Methods

1. Write a Python program using `plot()`.
2. Write a Python program using `title()`.
3. Write a Python program using `xlabel()`.
4. Write a Python program using `ylabel()`.
5. Write a Python program using `show()`.
6. Write a Python program using `grid()`.

7. Write a Python program using legend().
8. Write a Python program using xlim() and ylim().

34. Line Chart

1. Write a Python program to draw a line chart.
2. Write a Python program to plot monthly temperature using a line chart.
3. Write a Python program to plot marks of a student using a line chart.

35. Pie Chart

1. Write a Python program to draw a pie chart.
2. Write a Python program to create a pie chart of fruits.
3. Write a Python program to use labels, autopct, explode, shadow, and startangle.

36. Bar Chart

1. Write a Python program to draw a bar chart.
2. Write a Python program to compare marks using bar chart.
3. Write a Python program to customize bar chart colors.
4. Write a Python program to add legend in a bar chart.
5. Write a Python program to draw a bar chart with multiple colors.

37. Animation

37.1 Animation

1. Write a Python program related to animation concept using graphics or movement.
2. Write a program showing the movement of an object.

37.2 Types of Animation

1. Write a Python program or code-related question showing 2D animation concept.
2. Write a Python program or code-related question showing 3D animation concept.